

POWER STATION OPERATIONS TERMINAL

OFFICIAL OPERATIONS MANUAL - RIVERSIDE COAL POWER STATION

Keep the lights on. Keep your job.

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1. WELCOME TO THE JOB

It is 6:00 AM on a Monday morning.

You are sitting in the control room of the Riverside Coal Power Station, watching steam pressure gauges and listening to the hum of turbines.

Three months ago you were an IT consultant. You had a comfortable office job, spent your days in meetings about "synergising cross-functional paradigms," and your biggest worry was whether the coffee machine would be fixed by noon. Then came the conference call. Your manager, Karen, with her MBA from a school you had never heard of, decided your entire department could be "optimised" by outsourcing to an AI chatbot that cost twelve dollars a month.

After weeks of sending resumes into the void you finally got a call back. Not from a tech company. From Earl. Earl was the night-shift operator at Riverside for 37 years. He was three weeks from retirement when he decided to test whether the plant could run itself while he took a nap. It could not. The resulting incident involved a minor steam leak, the evacuation of two city blocks, and Earl's immediate retirement without the gold watch he had been promised.

The plant needed a new operator. Desperately. Your resume mentioned you "worked with computers" and "solved complex problems." Close enough.

Your training lasted exactly four hours. The plant supervisor - a grizzled man named Chuck who smells permanently of diesel and disappointment - walked you through the control room once.

He pointed at the pressure readings: *"Keep it around 160 bar. Too Low, no power. Too high, boom."*

He pointed at the drum level gauge: *"Keep water between 40% and 60%. Too low, tubes overheat. Too high, turbine gets wet. Wet turbines are expensive. Don't make the turbine wet."*

He gestured at the pulverizers: *"These grind coal into dust. They vibrate. Sometimes too much. If one trips, start another. We have four. Try to keep at least two running."*

Then Chuck handed you a coffee-stained operations manual from 1987, said *"You'll figure it out,"* and left to deal with a situation in Unit 2.

SUPERVISOR'S NOTE: Read the gauges. Press the buttons. Don't panic.

2. HOW A COAL POWER PLANT WORKS

You are responsible for a 300 MW coal-fired power station. Understanding what you are operating will help you keep it alive. This is the short version. The 1987 manual has 400 pages of the long version. Nobody reads the long version.

2.1 STAGE 1: COAL SYSTEM - FUEL PREPARATION

Raw coal arrives by conveyor and is fed into four **PULVERIZERS** - large grinding mills that reduce it to a fine powder. Pulverised coal is blown into the boiler furnace by primary air fans where it ignites and burns at approximately 1400°C. More coal burning = more heat = more steam. You control coal burn rate by adjusting each pulverizer's load from 0% to 100%. Overloading them causes vibration and motor overheating.

2.2 STAGE 2: BOILER - STEAM GENERATION

The boiler is a large vessel surrounded by water tubes. Heat from the burning coal converts that water to high-pressure steam. The target conditions are 160 bar pressure and 530°C temperature. A drum at the top of the boiler maintains the water level - this must stay close to 50%. Too low and the tubes overheat and rupture. Too high and water carryover damages the turbine. The **FEEDWATER SYSTEM** continuously replaces the water that boils off.

2.3 STAGE 3: TURBINE - MECHANICAL ENERGY

High-pressure steam from the boiler flows through the **MAIN STEAM VALVE** into the turbine. The steam expands across multiple blade stages, spinning the turbine shaft at 3000 RPM (50 Hz). The shaft drives the **GENERATOR** which produces the electrical output. Opening the steam valve increases power; closing it reduces power. The turbine responds to valve changes within 5-15 seconds.

2.4 STAGE 4: COOLING SYSTEM - HEAT REJECTION

Steam exhausted from the turbine enters the **CONDENSER**. Three **CIRCULATION PUMPS** draw cool water from the river through the condenser, removing heat and condensing the steam back into water. This creates a vacuum (0.046 bar) in the condenser - the lower the vacuum pressure, the more efficiently the

turbine extracts energy from the steam. Condensed water is returned to the boiler by the feedwater pumps, completing the cycle.

2.5 YOUR JOB

Balance all four stages to match the power demand issued by the grid operator (TSO) while keeping the plant within safe limits. Equipment will degrade. Faults will occur. The TSO does not care about any of that. Your job is to keep the lights on.

3. GETTING STARTED

3.1 SYSTEM REQUIREMENTS

MINIMUM CONFIGURATION:

OS : Windows 10 or later
PYTHON : 3.12 or later (source version)
DISPLAY : 960 x 540 minimum resolution
MEMORY : 512 MB RAM
STORAGE : 100 MB

3.2 LAUNCHING THE GAME

Executable version: Double-click COALCOM.exe

The game auto-detects your monitor resolution and selects the best display preset. Press **[ALT+ENTER]** at any time to toggle fullscreen.

3.3 MAIN MENU OPTIONS

NEW GAME	Start Campaign
LOAD GAME	Resume saved
CONTINUOUS MODE	Sandbox play
OPTIONS	Settings
EXIT	Leave

4. GAME MODES

4.1 CAMPAIGN MODE

Campaign is the main game. You work through 10 shifts of increasing difficulty at Riverside Power Station. Equipment health persists between shifts - a fault you mismanage today will cost you next shift. Complete all 10 shifts to finish the campaign.

SHIFT PROGRESSION:

Shift	Name	Duration (min)	Difficulty
1	First Day	3	ROOKIE
2	Morning Routine	3.5	ROOKIE
3	Afternoon Peak	5	ROOKIE
4	Load Cycling	7	OPERATOR
5	Equipment Stress	8	OPERATOR
6	Maintenance Deficit	12	OPERATOR
7	Grid Emergency	12	EXPERT
8	Cascading Failures	14	EXPERT
9	Perfect Storm	14	EXPERT
10	The Gauntlet	16	PROFESSIONAL

SUPERVISOR'S NOTE: Shifts 1 and 2 have no faults. Consider them practice. Shift 3 introduces your first equipment failure. Shift 10 will test everything you know. Good luck. You will need it.

CONTROL UNLOCK PROGRESSION:

Controls are introduced gradually to avoid overwhelming new operators.

Shift	Controls Unlocked	Page
1	Main Steam Valve [S]	F1
2	+ Feedwater Valve [W]	—
3	+ FW Pumps [Q][A]	F4
4	+ Coal Pulverizers [0-4] [R]	F2
5+	+ Circ Pumps [Z][X][C] + Backwash [B]	F3

EQUIPMENT HEALTH:

All seven major systems (BOILER, TURBINE, COAL, FEEDWATER, COOLING, CONTROL, ELECTRICAL) carry degraded health into the next shift. Health is tracked as **Equivalent Operating Hours (EOH)** --- the higher the EOH, the lower the health. Poor management and excessive start/stop cycles accelerate EOH accumulation. Low health increases the probability and severity of faults.

MAINTENANCE SYSTEM (spending available from Shift 5 onward):

Maintenance points are earned from Shift 1. From Shift 5 onward, you may spend them to reduce equipment EOH before the next shift. Each maintenance tier reduces EOH uniformly across *all* components. Maintenance is not available if the plant tripped.

Level	Cost (pts)	EOH Reduction
Basic	1	−200 EOH per component
Standard	4	−1,000 EOH per component
Full Overhaul	10	−3,000 EOH per component
Skip	0	No change

Points earned per grade: A = +2 pts, B = +1 pt, C = 0 pts, D = −1 pt, F = −3 pts (penalty).

4.2 CONTINUOUS MODE

Continuous Mode unlocks after completing Campaign Shift 5. It provides unrestricted sandbox play with full configuration control. There is no progression or equipment persistence - it is a training and practice environment.

CONFIGURATION OPTIONS:

Difficulty	Rookie / Operator / Expert / Professional
Faults Enabled	ON/OFF
Equipment Health	NEW / GOOD / WORN / POOR / CRITICAL
Control Preset	All / Tutorial / Progressive / Custom
Autopilot	ON/OFF

TRAINING SCENARIOS:

#	Name	Duration (min)	Faults
1	Morning Ramp	10	1
2	Afternoon Peak	15	3
3	Night Shift	12	3
4	Storm Surge - Rapid Response	10	3
5	Monday Morning	15	4
6	Friday Afternoon	12	4
7	Summer Heat Wave	18	7

Press [E] at any time during a Continuous Mode session to end the shift early.

SUPERVISOR'S NOTE: The demo limits campaign to Shift 4 and caps Continuous Mode sessions at 20 minutes. All features are otherwise available.

5. THE CONTROL ROOM

Your control room has five information pages accessible at any time with the function keys. A KEY INDICATORS BOX in the upper-right corner of every page always shows the most critical plant parameters.

KEY INDICATORS BOX (ALL PAGES):

POWER: 247 MW PRESSURE: 163 bar TEMP: 520 oC DRUM: 50 % COAL: 13 kg/s

Five numbers. Always visible. Always honest. If any of them is heading in the wrong direction, something needs your attention now --- regardless of which page you are on. Target: POWER matches the TSO order; PRESSURE 155--165 bar; TEMP 520--535°C; DRUM 40--60%; COAL appropriate for current load.

The plant STATUS banner below the box reads NORMAL, WARNING, ALARM, or TRIPPED. If it is not NORMAL, find out why before doing anything else.

5.1 [F1] MAIN OPS

Your home page. You will run most of your shift from here without switching away.

TSO ORDER panel (top left) --- current target MW, your actual output, and whether you are IN RANGE. This is the only number the TSO cares about. Watch the countdown; if it starts dropping fast, stop fine-tuning and hit the target.

BOILER panel (top centre) --- pressure, drum level, and steam temperature at a glance. Three numbers that tell you whether the boiler is happy. If pressure is drifting and you have not touched the coal load, suspect a leak.

FEEDWATER and COOLING panels (bottom row) --- pump status and valve position. If a pump shows STOPPED unexpectedly, act immediately. Drum level will not wait.

PLANT SCHEMATIC (right column) --- a live diagram of the full plant. Running equipment is highlighted; failed equipment goes dark. Use it to confirm your mental model when multiple things are happening at once.

5.2 [F2] COAL SYSTEM

(Unlocked at Shift 4.)

Four pulverizers, one table. Switch here the moment vibration or temperature misbehaves on the main page.

Watch: VIBRATION column. Green is fine; approaching 60 means reduce load; 80 is an alarm; 95 trips the mill automatically. MOTOR TEMP: alarm at 85°C, trip at 92°C. Catch either early and shed load before the mill forces your hand.

Summary line at the bottom shows total coal output (kg/s). Cross-check this against the COAL figure in the Key Indicators Box --- if they disagree, a feeder jam may have partially cut output.

5.3 [F3] COOLING SYSTEM

(Unlocked at Shift 5.)

Cooling is invisible until it goes wrong, then it quietly destroys your efficiency rating.

Condenser panel (left) --- vacuum and fouling percentage are the two numbers to watch. Fouling above 30% means run a backwash [B]. A degraded vacuum costs you turbine efficiency; at full degradation you lose around 7% output for the same fuel burn.

Circ Pump table (right) --- three pumps, bearing temperature, and vibration. Normally A and B run; C is standby. If a pump trips, start C immediately --- do not wait for the tripped pump's 20-second cooldown.

5.4 [F4] WATER/STEAM CYCLE

(Unlocked at Shift 3.)

The detailed view of everything between the drum and the turbine. Go here when the drum level is misbehaving and [F1] does not give you enough information.

Feedwater Pump table (top) --- status, load, bearing temperature, and vibration for both pumps. Pump A shows no power reading because it is turbine-driven. If bearing temperature on either pump is climbing toward 80°C, back off its load or prepare the other pump.

Boiler Heat Balance (bottom) --- coal feed in versus steam out. Any active tube leak appears here as unexplained steam loss. If the numbers do not add up, you have a leak.

5.5 [F5] EQUIPMENT HEALTH

Check this at the start of each shift and after any fault. It tells you how much trouble is waiting for you.

Systems Health (left) --- bar graphs for Boiler, Coal, Feedwater, Cooling, and Turbine. A system below 50% health will start generating faults with increasing frequency. Plan your load accordingly.

EOH table (centre) --- Equivalent Operating Hours per component. Higher EOH means higher fault probability. This is what maintenance spending reduces between shifts.

Shift Statistics (right) --- TSO order results, alarm and fault counts, fuel efficiency, and current auxiliary load. Your end-of-shift grade is calculated from these numbers. Check efficiency mid-shift if you are running heavy on coal.

5.6 ALARMS & MESSAGES PANEL (ALL PAGES)

Six-line event log at the bottom of every page. GREEN = information; YELLOW = TSO order expired; RED = alarm, fault, or trip.

Alarms freeze your attention until you acknowledge them with [SPACE]. Do not acknowledge blindly --- read the message first, identify the page you need ([F2] for coal, [F3] for cooling, [F4] for feedwater and boiler), then act. Acknowledging an alarm does not fix the underlying problem.

6. KEYBOARD REFERENCE

6.1 CONTROL FOCUS SELECTION

Key	Function
[S]	Select MAIN STEAM VALVE for adjustment
[W]	Select FEEDWATER VALVE for adjustment
[0]	Select ALL PULVERIZERS simultaneously
[1]	Select PULVERIZER A only
[2]	Select PULVERIZER B only
[3]	Select PULVERIZER C only
[4]	Select PULVERIZER D only

6.2 VALUE ADJUSTMENT

Key	Function
[UP]	Increase selected control by 1 step
[DOWN]	Decrease selected control by 1 step
[PAGE UP]	Increase selected control by 5 steps
[PAGE DN]	Decrease selected control by 5 steps

Step size: $\pm 0.5\%$ for valves, $\pm 1\%$ for pulverizer load.

6.3 PULVERIZER OPERATIONS

Key	Function
[R]	TOGGLE selected pulverizer (START if stopped, STOP if running)

Select a pulverizer with keys [1]-[4] then press [R] to start or stop. Ramp time: 30 seconds to full speed or full stop. A tripped pulverizer resets automatically after its cooldown expires --- select it and press [R] to restart.

6.4 FEEDWATER PUMP CONTROL

Key	Function
[Q]	Toggle FEEDWATER PUMP A (ON / OFF)
[A]	Toggle FEEDWATER PUMP B (ON / OFF, backup)

Pumps ramp over 10 seconds. Log shows: STARTING → RUNNING or STOPPING → STOPPED.

6.5 COOLING / CIRCULATION PUMP CONTROL

Key	Function
[Z]	Toggle CIRC PUMP A (ON / OFF)
[X]	Toggle CIRC PUMP B (ON / OFF)
[C]	Toggle CIRC PUMP C (ON / OFF, backup)
[B]	Start BACKWASH CYCLE (clears intake screens)

Pumps ramp over 20 seconds. Backwash cycle takes 90 seconds.

6.6 PAGE NAVIGATION

Key	Function
[F1]	MAIN OPS - Primary dashboard (always active)
[F2]	COAL SYSTEM - Pulverizer detail (Shift 4+)
[F3]	COOLING SYSTEM - Condenser / pumps (Shift 5+)
[F4]	WATER/STEAM CYCLE - Boiler / FW (Shift 3+)
[F5]	EQUIPMENT HEALTH - System health status (all shifts)

6.7 SYSTEM CONTROLS

Key	Function
[ESC]	Pause game / open pause menu
[SPACE]	Acknowledge active alarm
[I]	Show / hide in-game instruction manual
[E]	End shift (Continuous Mode only)
[ALT+ENTER]	Toggle fullscreen

6.8 PAUSE CONTROLS

Key	Function
[ESC]	Resume game
[S]	Save game (Campaign only)
[Q]	Quit to main menu
[I]	Show in-game manual

7. PLANT SYSTEMS

7.1 MAIN STEAM VALVE

The primary power control. Opening the valve allows more steam to reach the turbine; closing it reduces flow. Power responds within 5-15 seconds. This is the first control you learn and the one you will use most often.

Select: [S] Adjust: [UP] / [DOWN] ($\pm 0.5\%$ per step) Range: 0% to 100%
Effect: ± 0.5 -2 MW per 1% valve change.

SUPERVISOR'S NOTE: Make small, deliberate adjustments. A 10% lurch on the steam valve will upset the boiler pressure and drum level simultaneously. Chuck calls this "the domino effect." It is not complimentary.

7.2 BOILER SYSTEM

The boiler converts water to high-pressure steam. Three parameters demand constant attention: pressure and drum level.

Parameter	Target	Warning	Alarm	Trip
Steam Pressure	160 bar	<128 bar	>192 bar	>228 bar or <50 bar
Drum Level	50%	<40%	>60%	>95% or <15%

7.3 FEEDWATER SYSTEM

Water must be continuously supplied to replace what boils off. Two pumps provide redundancy: [Q] Pump A is turbine-driven (turbo-pump --- it runs from HP steam extraction and draws no electrical power). [A] Pump B is motor-driven and consumes 4.0 MW of auxiliary power. Prefer Pump A in normal operation; use Pump B as backup or when A is unavailable. Nominal flow: 265 kg/s at 100% valve.

Both pumps are monitored for bearing temperature (alarm 80°C, trip 88°C) and vibration (warn 55%, alarm 75%, auto-trip 90%). Monitor the [F4] pump table for early warning signs. After a pump trips, a 10-second cooldown is required before restart.

SUPERVISOR'S NOTE: If the drum level drops below 40%, open the feedwater valve. If it rises above 60%, close it. If both pumps fail simultaneously, pray. Then reduce load immediately and start the backup.

7.4 COAL PULVERIZERS (A, B, C, D)

Four mills grind coal to powder and deliver it to the furnace. Total coal burn rate determines the boiler's heat input and therefore the maximum achievable steam output and power.

Parameter	Warning	Trip
Mill Load	>90%	N/A
Vibration	>60	>95
Motor Temp.	>85°C	>92°C

Motor power draw: up to 0.5 MW per pulverizer at 100% load (scales with load %). Running all four at full load consumes ~2 MW of auxiliary load.

7.5 COOLING SYSTEM

Three circulation pumps move river water through the condenser to remove heat from the exhaust steam. Adequate cooling maintains the condenser vacuum - poor vacuum reduces turbine efficiency significantly (-7% at fully degraded vacuum).

Pumps A [Z] and B [X] normally running. Pump C [C] standby. Backwash [B] clears river debris (90 sec). Target vacuum: 0.046 bar. Backwash recommended when fouling exceeds 25--30%. Each pump draws 2.5 MW of auxiliary electrical power when running. After stopping, a 20-second cooldown must elapse before a pump can restart.

7.6 TURBINE / GENERATOR

The turbine converts steam energy into shaft rotation at 3000 RPM, driving the generator. You do not directly control the turbine beyond the main steam valve. Nominal output: 300 MW. Efficiency improves with higher condenser vacuum.

8. TSO ORDERS - MEETING DEMAND

The Transmission System Operator (TSO) manages the national grid and issues power output targets to all generators. Your job is to match those targets within the compliance window before the order expires.

An order is COMPLETE when you maintain power within the tolerance band continuously for the required hold duration.

Difficulty	Tolerance	Hold Time	Expiry
Rookie	± 5 MW	20 sec	100 sec
Operator	± 5 MW	25 sec	90 sec
Expert	± 5 MW	30 sec	80 sec
Professional	± 4 MW	35 sec	70 sec

If the order timer runs out before you reach compliance, the order is marked EXPIRED and the next order is issued after a 15-second window. Expired orders reduce your TSO Compliance score.

9. SCORING AND GRADES

At the end of each shift your performance is assessed across three areas and combined into an overall grade:

50% TSO Compliance + 30% Equipment Management + 20% Fuel Efficiency

TSO Compliance --- percentage of grid orders successfully completed on time:

Grade	Orders completed
A	≥95%
B	≥85%
C	≥70%
D	≥50%
F	<50%

Equipment Management --- penalty points accumulated from alarms and faults:

Grade	Penalty points
A	0
B	1
C	2--3
D	4--5
F	6+

Fuel Efficiency --- energy produced per kg of coal burned, vs. the target rate:

Grade	Efficiency vs. target
A	≥95%
B	≥85%
C	≥70%
D	≥50%
F	<50%

Maintenance points are earned based on your overall grade: A = +2 pts, B = +1 pt, C = 0 pts, D = -1 pt, F = -3 pts.

SUPERVISOR'S NOTE: Tripping the plant is an automatic grade F regardless of TSO compliance or efficiency scores. Don't trip the plant.

10. FAULT REFERENCE

Equipment will fail. This is expected, budgeted for, and entirely your problem.

SUPERVISOR'S NOTE: Read this section. Memorise it if you can. Faults managed within 30 seconds rarely cause lasting damage. Faults ignored for two minutes regularly cause plant trips.

10.1 BOILER FAULTS

Fault	Severity	Symptoms / Response
Minor Tube Leak	MINOR	Steam loss 10 kg/s, pressure drop. Reduce load, increase feedwater. Permanent.
Major Tube Leak	MAJOR	Steam loss 20 kg/s, severe pressure drop. Emergency load reduction, max feedwater immediately. Permanent.
Superheater Rupture	CRITICAL	Steam loss 40 kg/s. Plant trip imminent. No recovery - automatic shutdown in seconds.
Economizer Fouling	MODERATE	15% turbine efficiency loss, rising fuel demand. Wait for automatic soot blowing cycle. Clears in ~45 sec.

10.2 COAL SYSTEM FAULTS

Fault	Severity	Symptoms / Response
Pulverizer Seizure	MAJOR	Mill stops immediately. Switch to remaining mills, reduce load. Permanent.
Coal Feeder Jam	MODERATE	Affected mill output drops 50%. Stop and restart pulverizer to clear jam. ~2 min.
Motor Overload	MODERATE	Pulverizer trips (load was >95% too long). Restart at lower load (<90%). 1 min cooldown.
Primary Air Fan Fail	CRITICAL	All pulverizers drop to 30% output. Immediate emergency load reduction. Permanent.

10.3 COOLING SYSTEM FAULTS

Fault	Severity	Symptoms / Response
Circ Pump Trip	MAJOR	Loss of cooling flow, vacuum degrades. Start backup pump within 30 seconds.
Intake Screen Block	MODERATE	30% cooling flow reduction, rising condenser temp. Press [B] for backwash (~3 min).
Condenser Tube Leak	MAJOR	10% efficiency loss, rising back-pressure. Reduce load. Permanent until maintenance.
Cooling Tower Fan	MODERATE	Cooling water temp rises +5°C, reduced heat rejection. Increase pump flow or reduce load. Permanent.

10.4 WATER / STEAM SYSTEM FAULTS

Fault	Severity	Symptoms / Response
Feedwater Pump Trip	CRITICAL	Drum level falling fast. START BACKUP PUMP IMMEDIATELY. Approx. 10 seconds before trip.
Steam Valve Stuck	MAJOR	Cannot adjust power output. Wait - valve auto-recovers within 5-15 seconds.
Attemperator Failure	MODERATE	Steam temperature rising uncontrolled. Lower firing rate (reduce pulverizer load). Resolves in ~4 min.

10.5 CONTROL / ELECTRICAL FAULTS

Fault	Severity	Symptoms / Response
Drum Level Sensor Malfunction	MODERATE	Drum level reading erratic / noisy. Trust the trend, not the exact number. Resolves in ~10 seconds.
Power Meter Malfunction	MODERATE	Power reading erratic / noisy. Use steam valve position as proxy for output. Resolves in ~10 seconds.

11. OPERATOR'S GUIDE - EFFICIENT OPERATION

11.1 POWER RAMPING STRATEGY

When a new TSO order arrives, resist the urge to slam the steam valve to the new target. The correct sequence is:

1. **Anticipate the load direction.** Is the new target higher or lower?
2. **Adjust the steam valve first.** This gives immediate power response.
3. **Adjust coal feed to match.** More power = more coal. Allow 10-15 seconds for coal burn changes to affect steam pressure.
4. **Balance feedwater.** Higher firing rate means more steam; open the feedwater valve slightly to maintain drum level.
5. **Wait and observe.** Let the plant stabilise before further adjustment. Power oscillations usually damp out in 15-30 seconds.

Rule of thumb: Move the valve, wait, then move coal. Never move both at maximum rate simultaneously.

11.2 PULVERIZER LOAD BALANCING

Running all four mills at 50% is safer than running two at 100%. Spread the load evenly across running mills to minimise vibration, reduce motor temperatures, and extend equipment life.

Watch vibration closely. A mill climbing toward 60 units should have its load reduced. A mill at 80 is alarming. At 95 it will trip automatically.

11.3 FEEDWATER PUMP STRATEGY

Normally only one pump needs to run. Run both when output is above 270 MW, one pump shows degradation warnings, or you anticipate a rapid load increase. If a pump trips unexpectedly, start the other immediately --- you have approximately 10 seconds before drum level drops to alarm. Pump A [Q] is turbine-driven (no electrical load --- always prefer it as primary). Pump B [A] is motor-driven and consumes 4.0 MW. Keep Pump A running whenever possible; switch to Pump B only when A has tripped or is in cooldown.

11.4 COOLING PUMP STRATEGY

Two circulation pumps (A and B) are normally running. Pump C is the backup. Never run fewer than two pumps at full load --- insufficient cooling raises condenser back-pressure and reduces turbine efficiency by up to 7%. Check fouling percentage on the [F3] page periodically; backwash when fouling exceeds 25--30%. Monitor bearing temperature (alarm at 80°C) and vibration on the pump table. If a pump trips, remember the 20-second cooldown before it can restart --- start Pump C immediately rather than waiting.

11.5 RESPONDING TO FAULTS

When an alarm appears: (1) note the fault type; (2) check the relevant page ([F2] coal, [F3] cooling, [F4] feedwater/boiler); (3) take the prescribed response from Section 10; (4) acknowledge the alarm [SPACE]; (5) monitor the outcome.

Most time-critical faults: Feedwater pump trip (10 sec to start backup); circ pump trip (30 sec to restore cooling); major tube leak (immediate load reduction).

11.6 READING TRENDS, NOT SNAPSHOTS

A drum level of 42% is not alarming. A drum level falling from 55% to 42% over the last 20 seconds is alarming. Always assess whether a value is stable, rising, or falling. A parameter near a warning threshold but stable is usually fine. A parameter heading toward the limit at speed requires intervention *now*, not when the alarm activates.

11.7 EFFICIENCY TIPS

Efficiency score is based on coal burned relative to electricity produced.

Avoid unnecessary load cycling --- combustion efficiency drops temporarily during rapid ramps. Match coal feed to actual demand: do not run four mills at 70% when two at 75% would produce the same output. Maintain good condenser vacuum. Avoid high drum level, as water carryover reduces steam quality and turbine efficiency.

Auxiliary load: pumps and mills draw from your own output before the meter reaches the grid. The [F5] Shift Statistics box shows the live total.

Equipment	Notes	Load (MW)
Feedwater Pump A	Steam-driven	---
Feedwater Pump B	Motor-driven	4.0
Circ Pump A	Per pump when running	2.5
Circ Pump B	Per pump when running	2.5
Circ Pump C	Per pump when running	2.5
Pulverizer (each)	At 100% load; scales with load %	0.5
<i>Worst case</i>	<i>FW-B + 3 circ + 4 pulv @ 100%</i>	<i>13.5</i>
<i>Best case</i>	<i>FW-A + 2 circ + 2 pulv @ 50%</i>	<i>5.5</i>

Prefer Pump A whenever possible. Run only the circ pumps you need. Keep pulverizer loads balanced and no higher than demand requires. The difference between worst and best case is nearly 8 MW --- enough to push a borderline TSO order into compliance.

SUPERVISOR'S NOTE: The operators who score A-grades are not faster than the others. They make fewer large corrections. Small movements, steady hands, read the trend. That is the whole secret. You are welcome.

12. ALARM REFERENCE

Quick reference for all alarm conditions and trigger thresholds.

Pressure Alarms (target: 160 bar):

Condition	Threshold	Action
Low Pressure Warning	<128 bar	Yellow log
High Pressure Warning	>192 bar	Yellow log
Low Pressure Trip	<50 bar	PLANT TRIP
High Pressure Trip	>228 bar	PLANT TRIP

Drum Level Alarms (target: 50%):

Condition	Threshold	Action
Low Level Warning	<40%	Yellow log
High Level Warning	>60%	Yellow log
Low Level Trip	<15%	PLANT TRIP
High Level Trip	>95%	PLANT TRIP

Pulverizer Alarms:

Condition	Threshold	Action
Vibration Warning	>60	Reduce load
Vibration Alarm	>80	Red log
Vibration Auto-Trip	>95	Mill trips
Bearing Temp Warning	>85°C	Yellow log
Bearing Temp Trip	>92°C	Mill trips

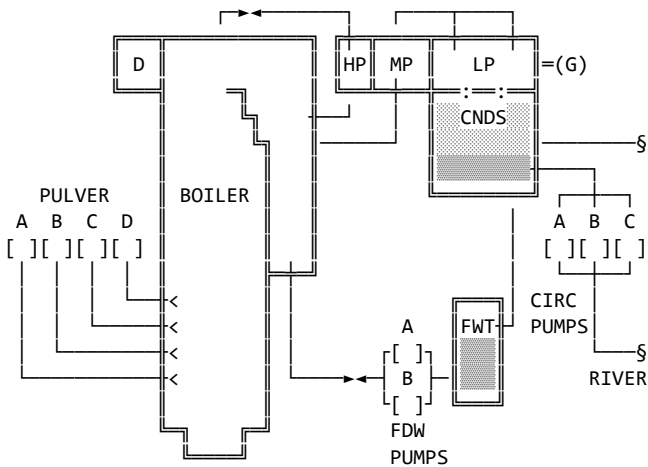
Pump Alarms (Feedwater and Circulation):

Condition	Threshold	Action
Vibration Warning	>55%	Reduce load or start backup
Vibration Alarm	>75%	Red log
Vibration Auto-Trip	>90%	Pump trips
Bearing Temp Warning	>80°C	Yellow log
Bearing Temp Trip	>88°C	Pump trips (motor pumps only)

13. QUICK REFERENCE

13.1 PLANT SCHEMATIC

The diagram below shows how the major plant systems connect. This schematic appears on every information page in the control room.

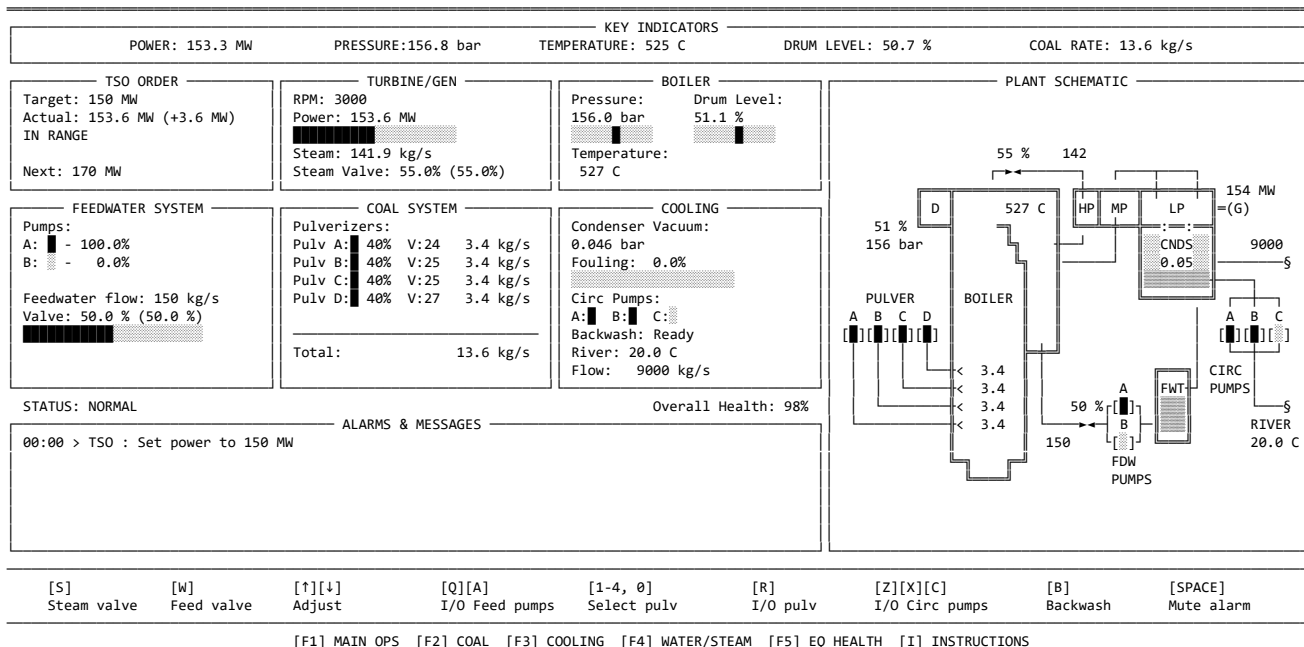


Symbol	Meaning
A B C D (left)	Coal pulverizers
D	Steam drum (top of boiler)
HP / MP / LP	Turbine stages: High / Medium / Low Pressure
(G)	Generator
CNDS	Condenser
A B C (right)	Circulating water pumps
§	River intake / discharge
FWT	Feedwater tank (deaerator)
A B (center)	Feedwater pumps (A = turbo-driven, B = motor-driven)

13.2 CONTROL ROOM SCREENS

The following pages show real control room screens captured during operation. Each page is shown at full width --- rotate the manual (or your PDF viewer) if reading in print.

[F1] MAIN OPERATIONS DASHBOARD

[CONTINUOUS MODE]
Time: 00:15COALCOM: POWER STATION OPERATIONS TERMINAL
Scenario: Morning Ramp

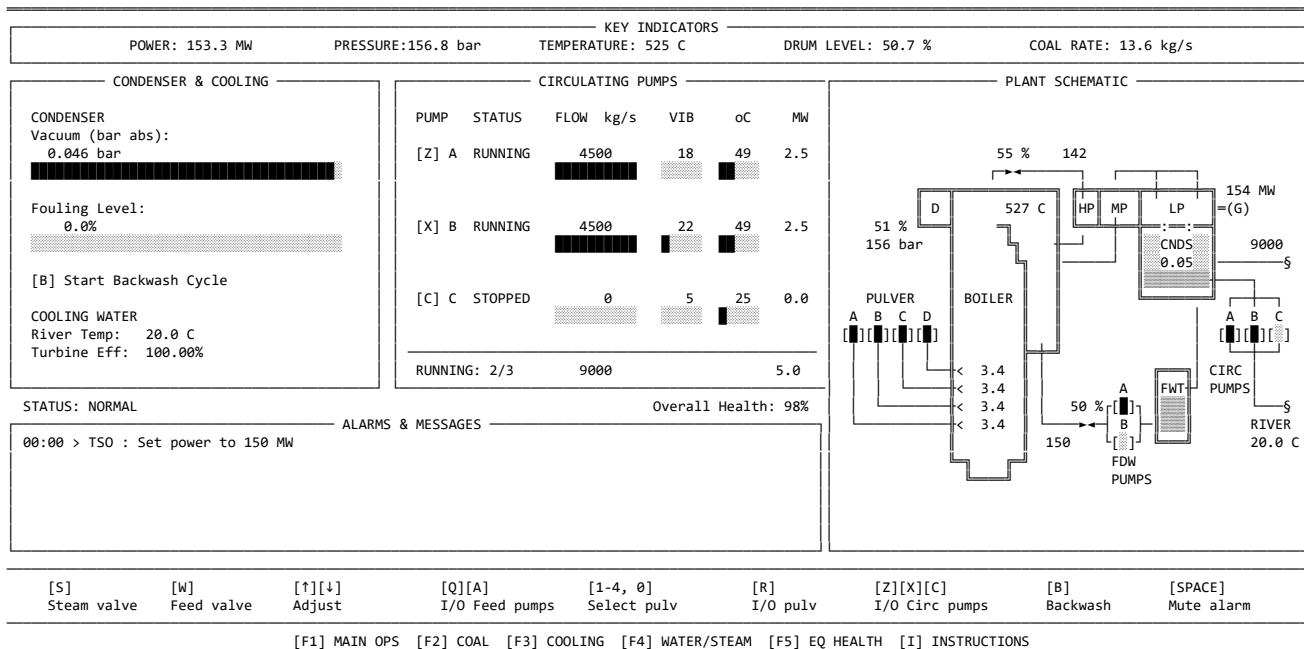
COALCOM OPERATIONS MANUAL

v1.0.0 / 1987

COALCOM: POWER STATION OPERATIONS TERMINAL
Scenario: Morning Ramp

COALCOM OPERATIONS MANUAL

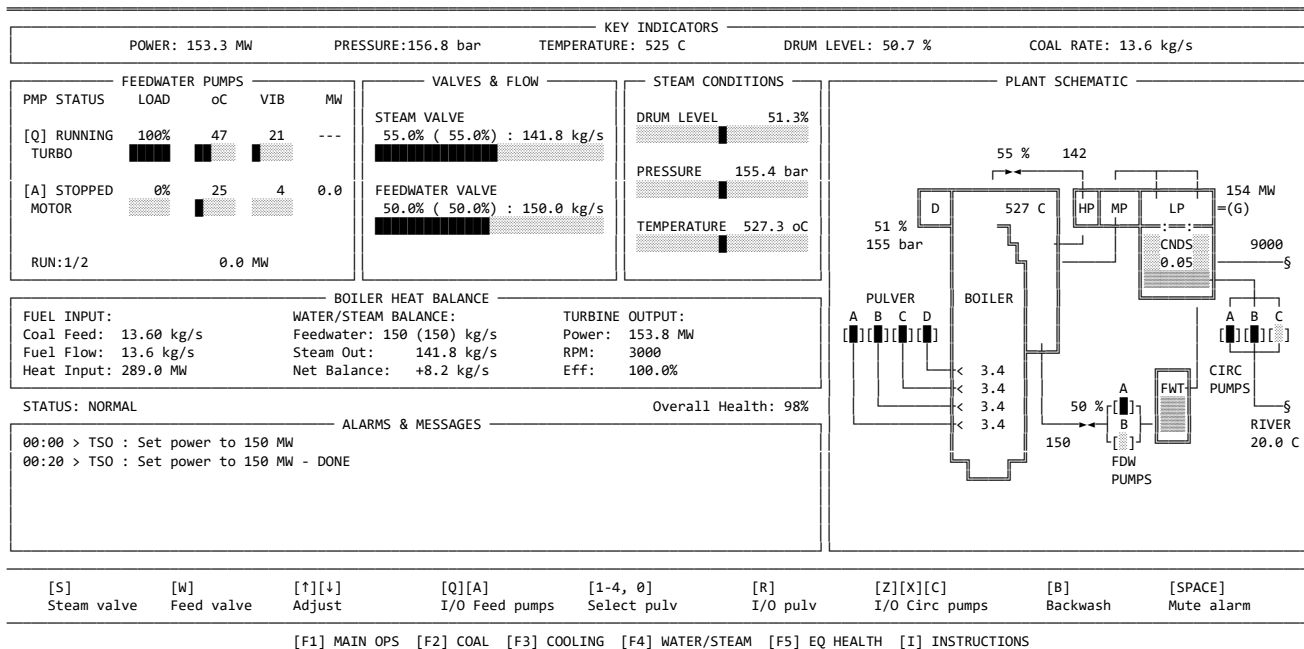
v1.0.0 / 1987

[F3] COOLING SYSTEM[CONTINUOUS MODE]
Time: 00:18COALCOM: POWER STATION OPERATIONS TERMINAL
Scenario: Morning Ramp

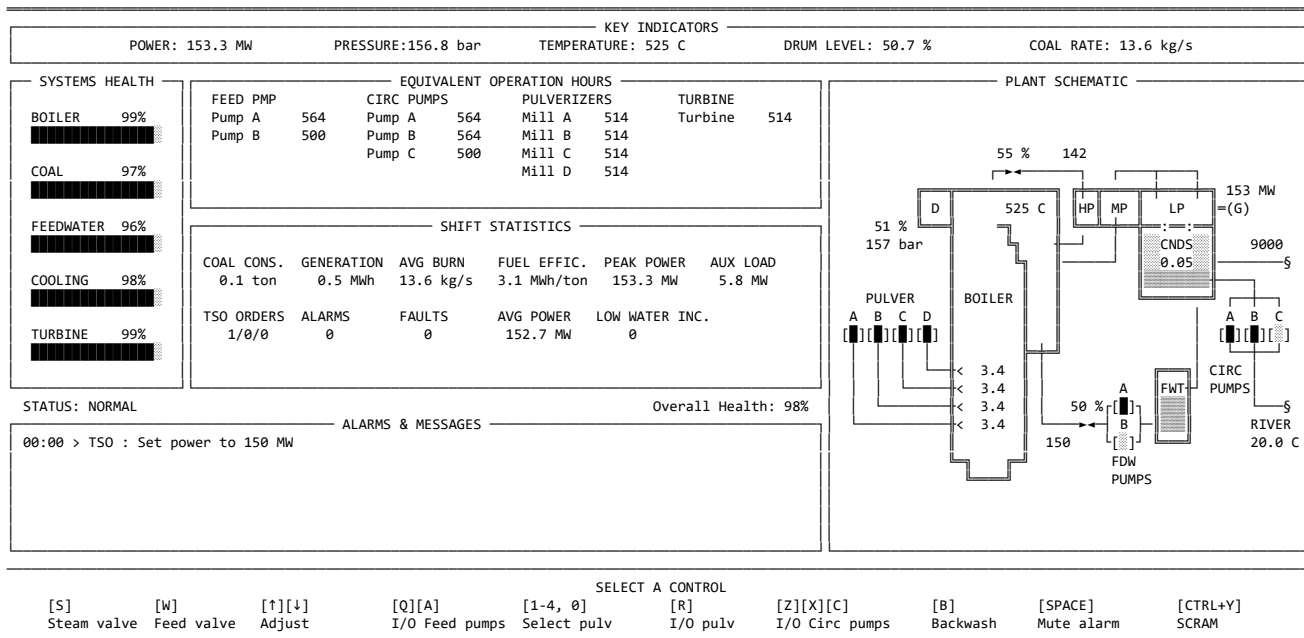
[F4] WATER/STEAM SYSTEM

[CONTINUOUS MODE]
Time: 00:20

COALCOM: POWER STATION OPERATIONS TERMINAL
Scenario: Morning Ramp



[F5] EQUIPMENT HEALTH & STATISTICS SYSTEM

[CONTINUOUS MODE]
Time: 00:10COALCOM: POWER STATION OPERATIONS TERMINAL
Scenario: Morning Ramp

SELECT A CONTROL

[S] Steam valve	[W] Feed valve	[↑][↓] Adjust	[Q][A] I/O Feed pumps	[1-4, 0] Select pulv	[R] I/O pulv	[Z][X][C] I/O Circ pumps	[B] Backwash	[SPACE] Mute alarm	[CTRL+Y] SCRAM
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[F1] MAIN OPS [F2] COAL [F3] COOLING [F4] WATER/STEAM [F5] EQ HEALTH [I] INSTRUCTIONS

COALCOM - POWER STATION OPERATIONS TERMINAL

Created by Pedro Matos @ 2026

Keep the lights on. Keep your job.

This manual is provided as-is. Riverside Power Station accepts no liability for plant trips, grid outages, loss of employment, or incidents involving Earl-style supervisory napping.

If in doubt: read the gauges, press the buttons, don't panic.